

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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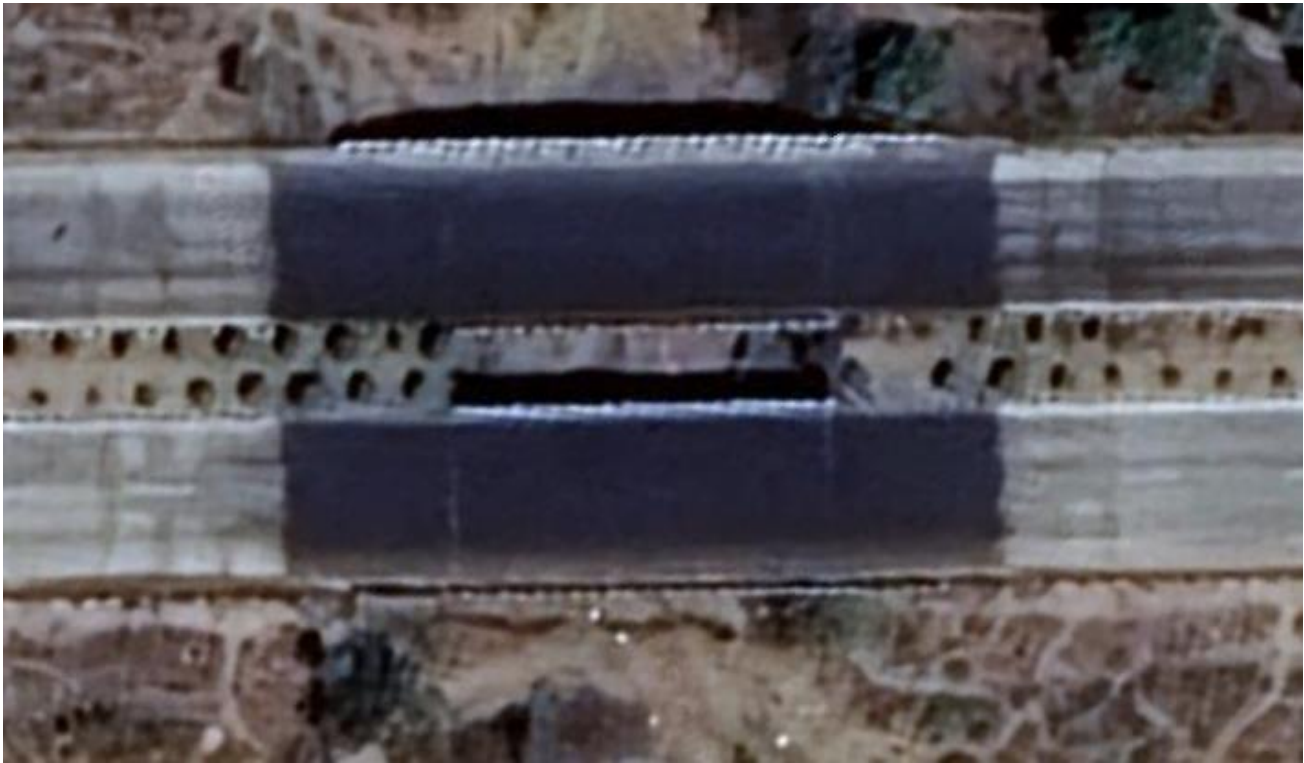
3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage –901+270

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

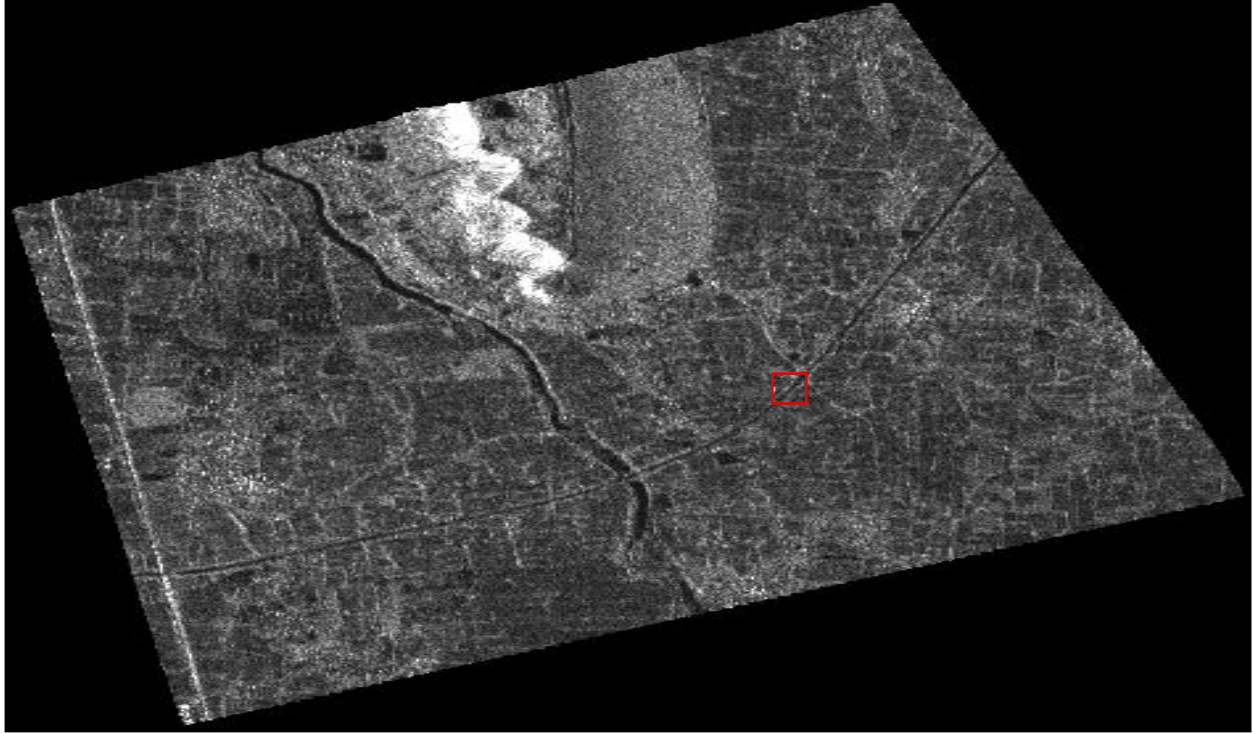
Flight Direction - DESCENDING

Polarization – VV+VH

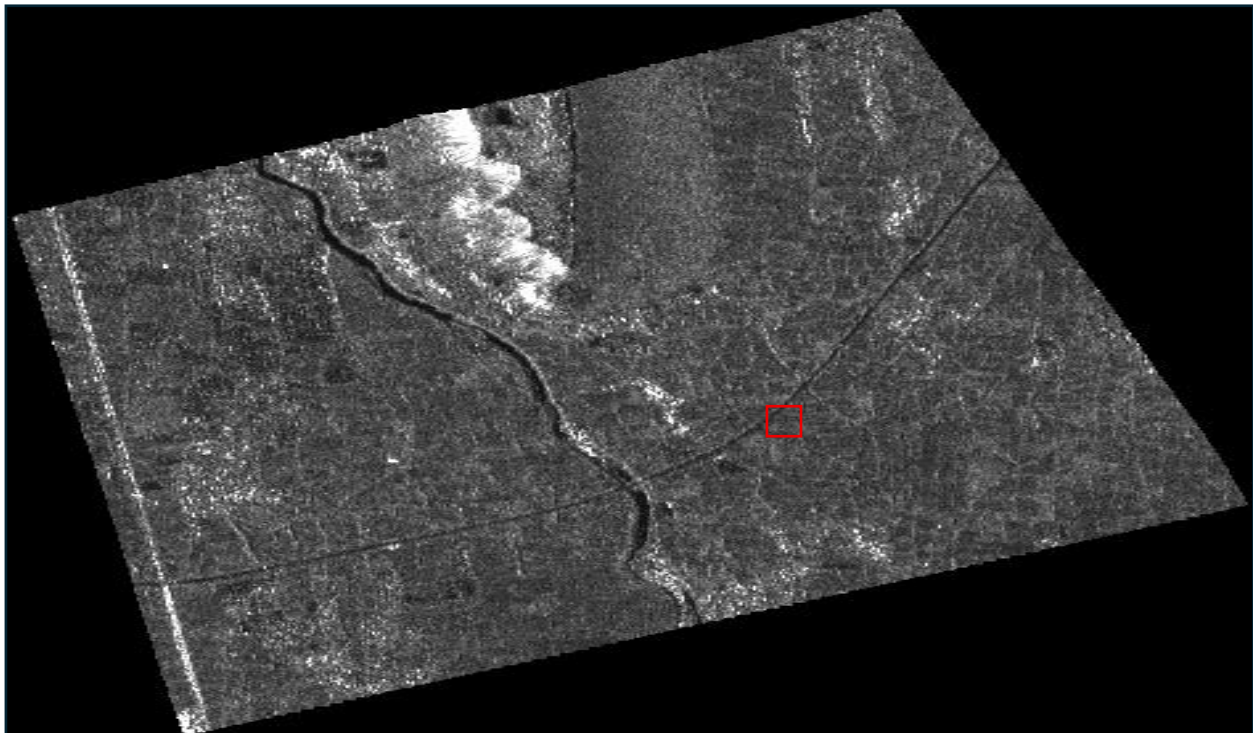
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

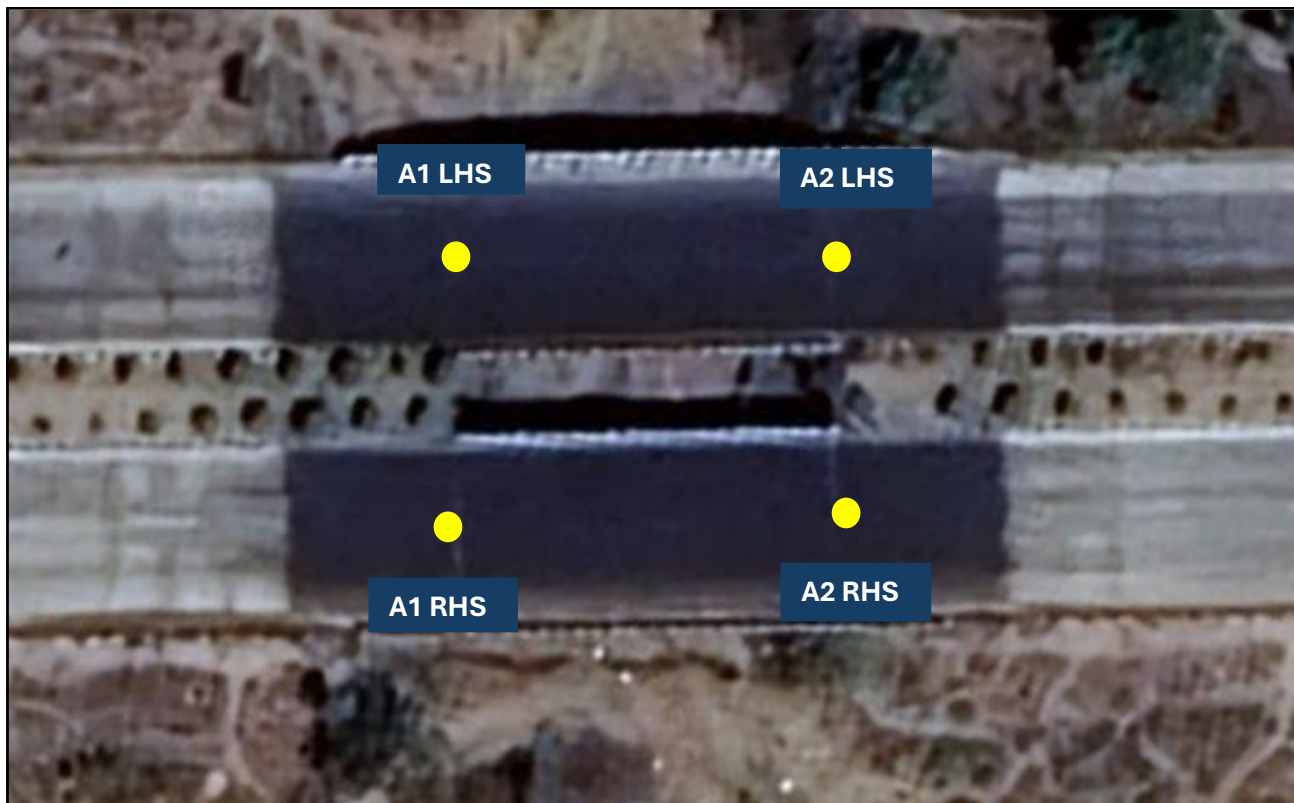


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

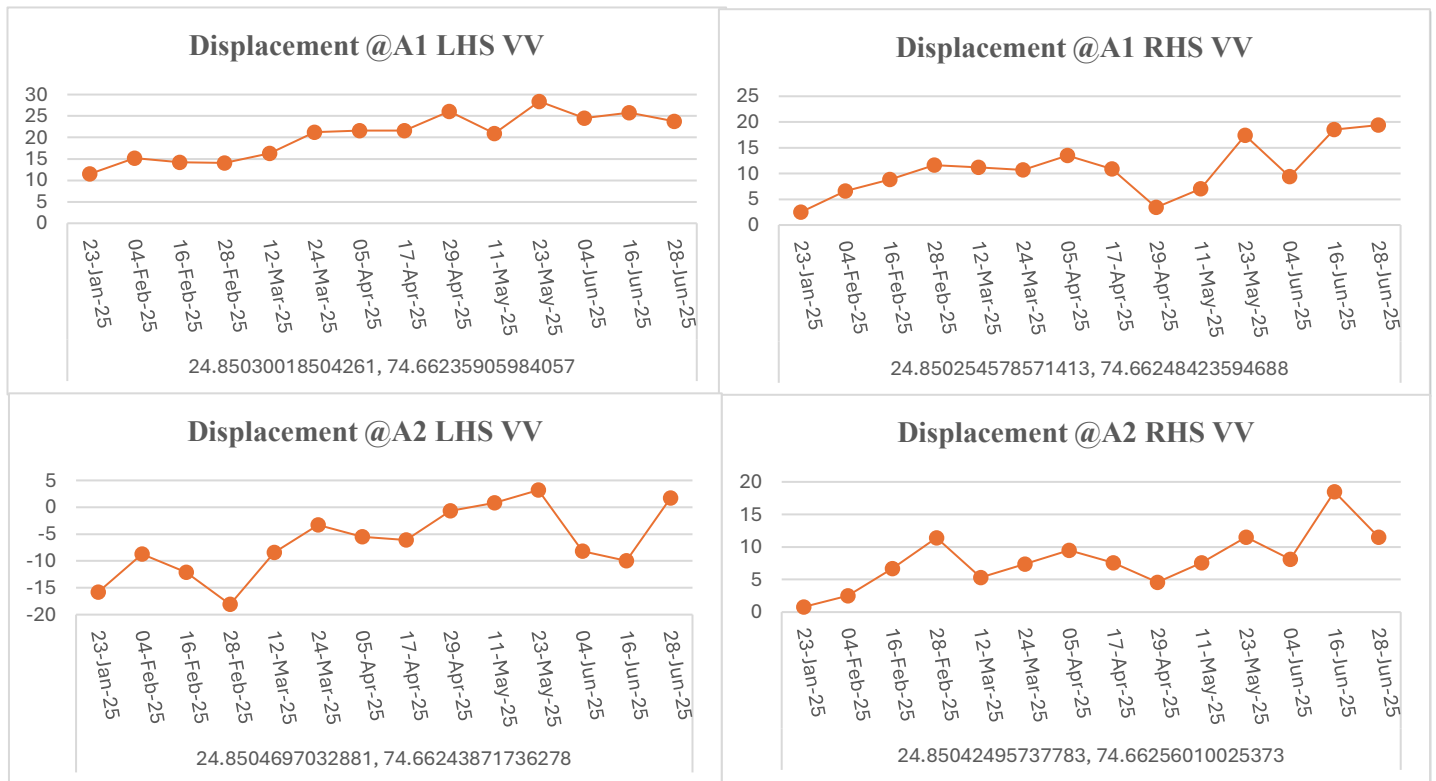
MNB – CH-901+270

Length – 37.92M

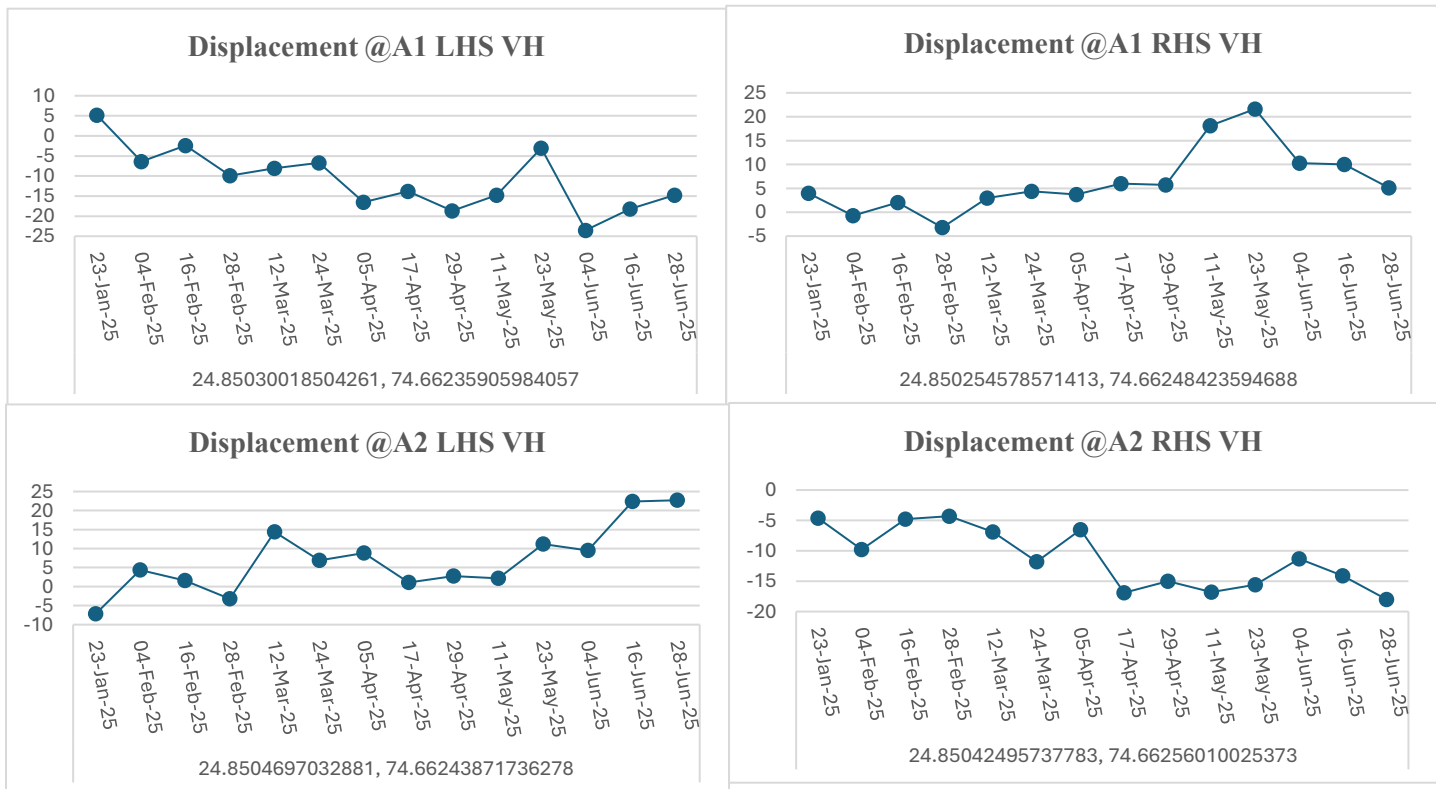


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

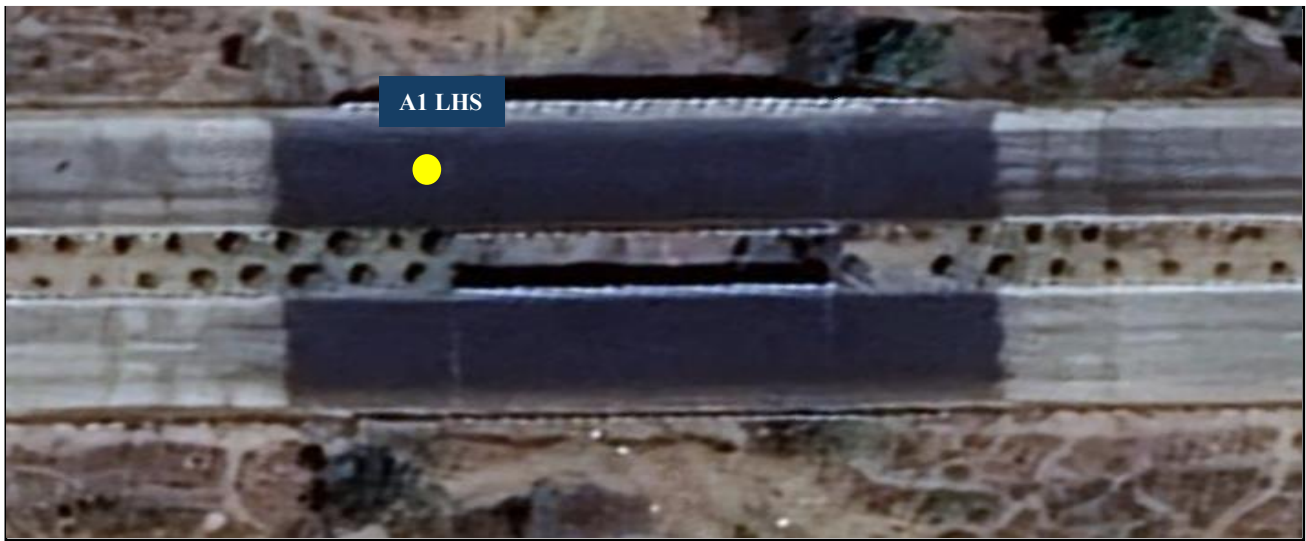
3.1 VV Displacement from 23 Jan 2025 To 28 June 2025



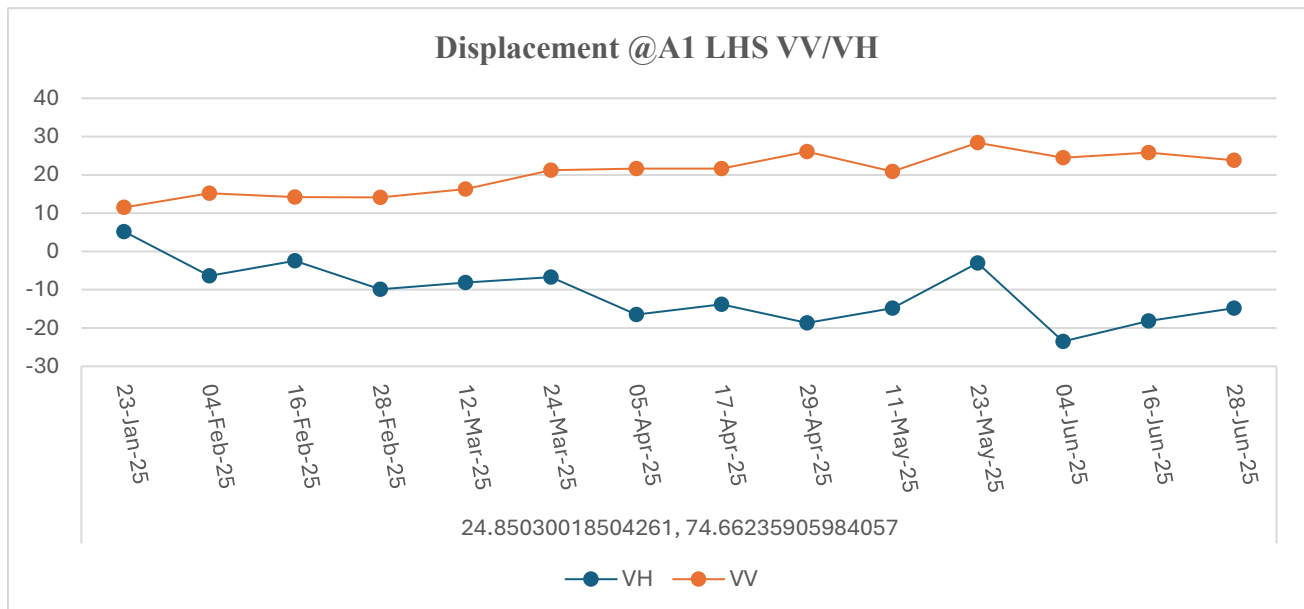
3.2 VH Displacement from 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.85030018504261, 74.66235905984057)



Key Observations:

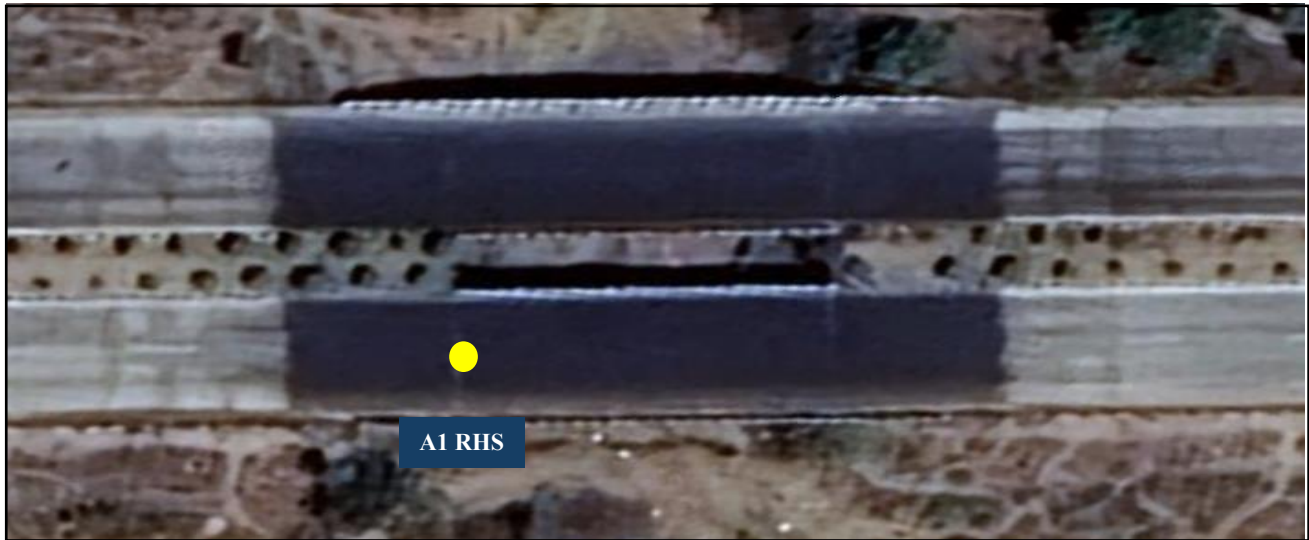
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV remains consistently positive, ranging approximately from +10 mm to +28 mm, indicating sustained vertical uplift.
- **Horizontal Displacement (VH)** VH displacement is predominantly negative throughout the monitoring period, ranging approximately from –25 mm to +5 mm.

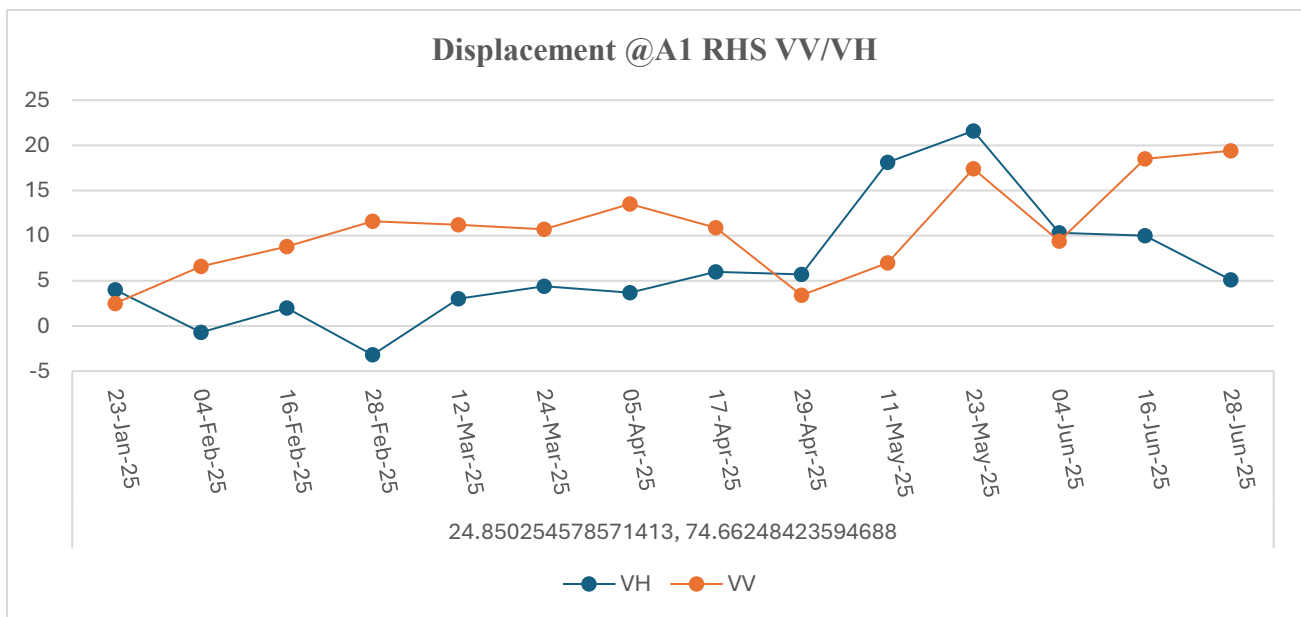
Critical Events & Anomalies:

- **April-May 2025:** A phase of simultaneous increasing uplift (VV) and deepening negative VH is observed, indicating coupled vertical–lateral structural response.
- **June 2025:** VH reaches its maximum negative excursion (–25 mm) while VV remains high, suggesting differential movement concentration at A1 LHS.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.850254578571413, 74.66248423594688)



Key Observations:

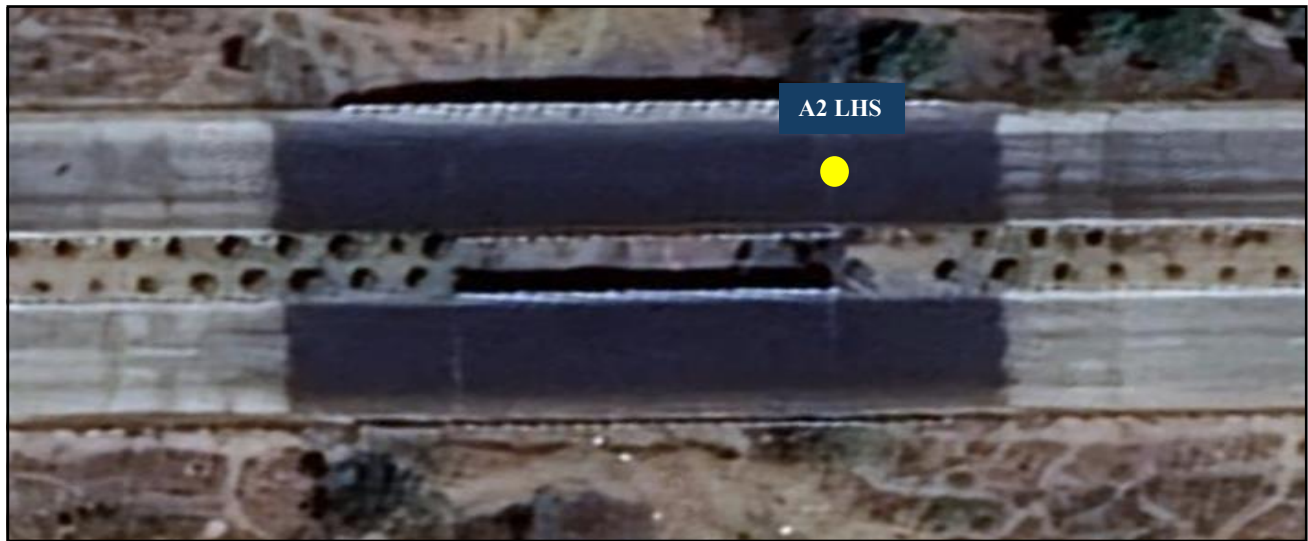
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH shows moderate variability, ranging approximately from –4 mm to +21 mm over the monitoring period.
- **Vertical displacement (VV)** VV remains consistently positive throughout, ranging approximately from +4 mm to +19 mm.

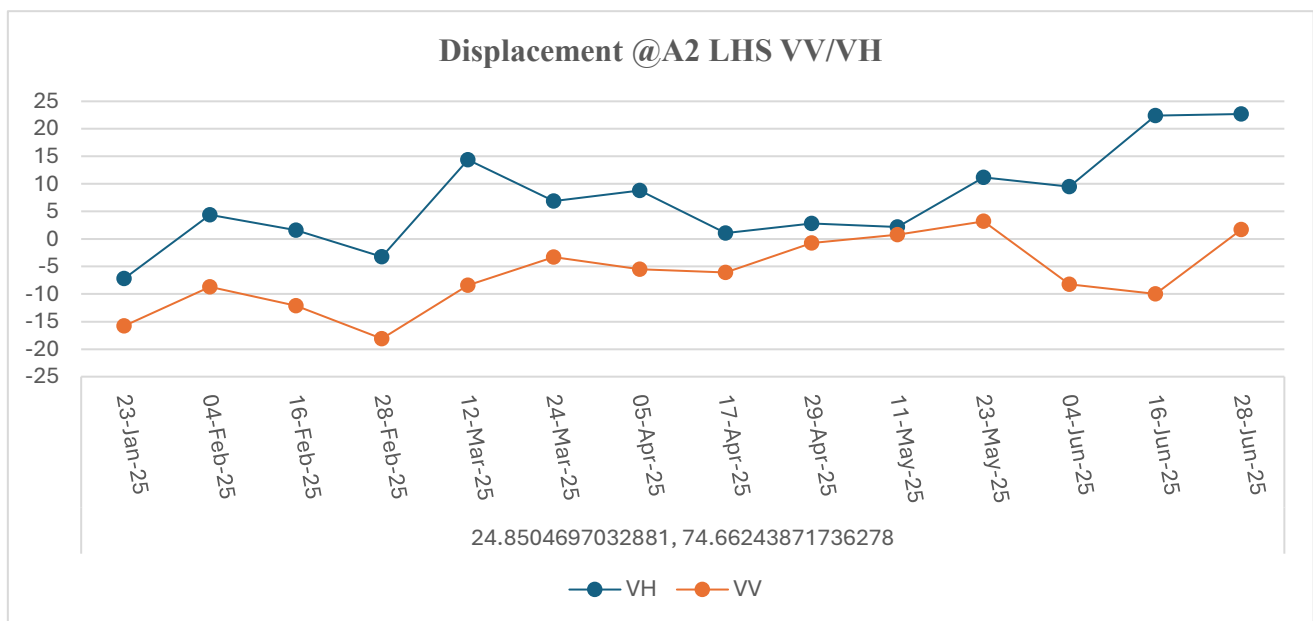
Critical Events & Anomalies:

- **May 2025:** Both VH and VV reach near-simultaneous peak values, indicating a period of maximum combined displacement demand at A1 RHS.
- **June 2025:** A short-term reduction in both components is observed, suggesting temporary unloading or thermal moderation, followed by stabilisation.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.8504697032881, 74.66243871736278)



Key Observations:

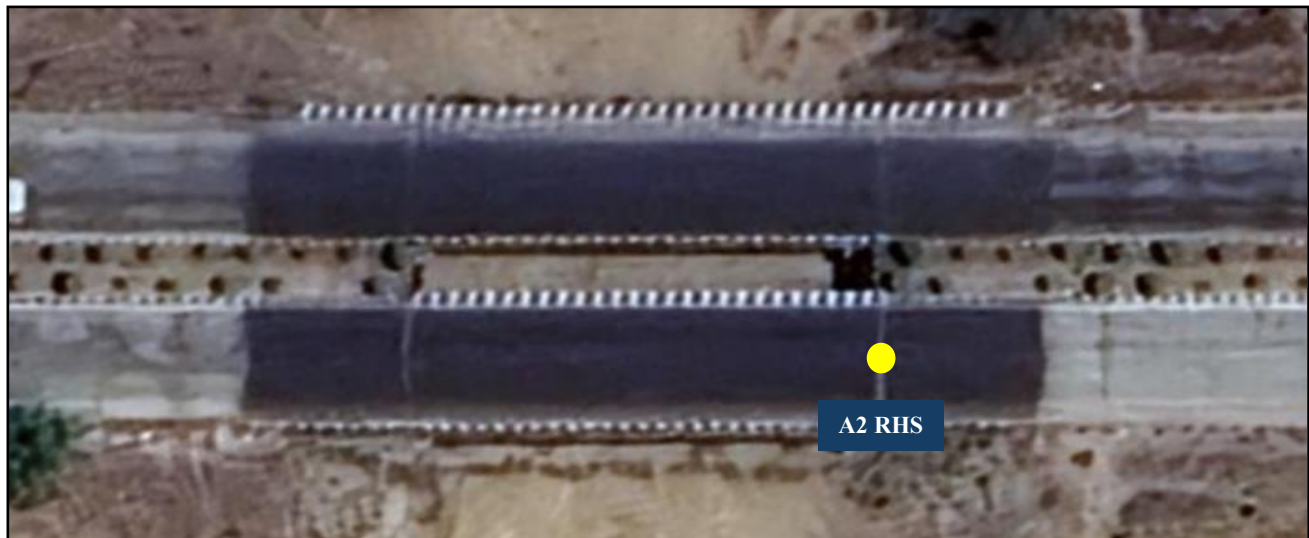
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH exhibits a progressive positive trend over the monitoring period, ranging approximately from –6 mm to +23 mm.
- **Vertical Displacement (VV)** VV remains predominantly negative, indicating net settlement, with values ranging approximately from –18 mm to +2 mm.

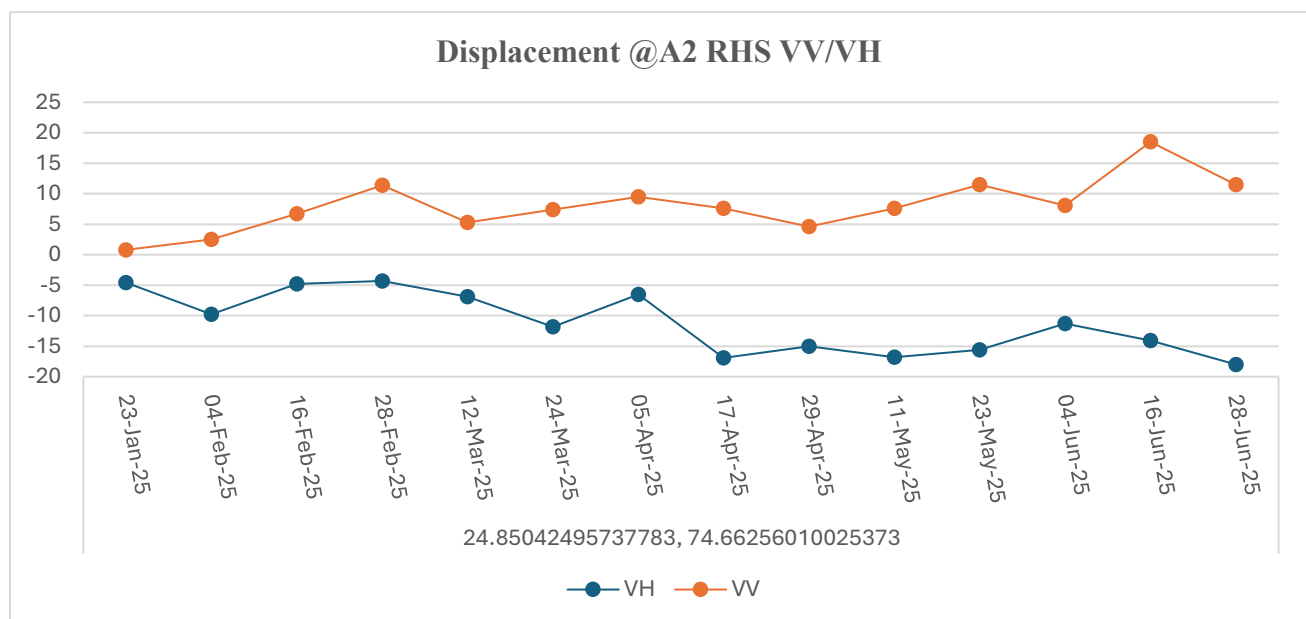
Critical Events & Anomalies:

- **March 2025:** A sharp increase in VH coincides with reduced settlement, implying a possible change in load path or boundary condition (e.g., temperature rise or traffic pattern shift).
- **June 2025:** Opposing trends are evident VH increases sharply while VV drops, indicating lateral demand dominance over vertical response.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.85042495737783, 74.66256010025373)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV values are predominantly positive, ranging approximately from +2 mm to +19 mm, indicating net uplift or upward vertical response.
- **Horizontal (VH) displacement** VH values remain consistently negative throughout the monitoring period, indicating a persistent lateral movement in the opposite direction to A2 LHS.

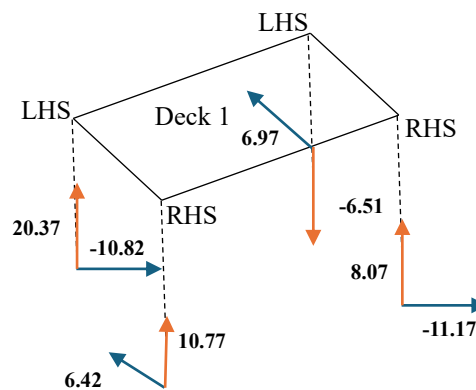
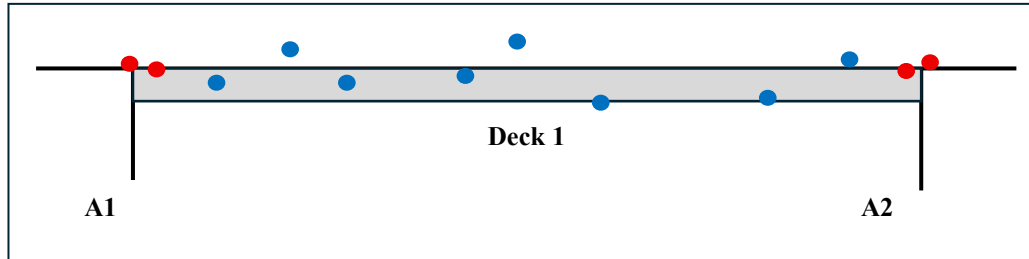
Critical Events & Anomalies:

- **March 2025:** A sharp divergence appears where VH becomes more negative while VV increases, suggesting combined lateral translation with vertical rotation effects at A2 RHS.
- **May-June 2025:** The simultaneous increase in uplift (VV) and negative VH drift indicates asymmetric abutment behaviour, potentially influenced by thermal expansion, bearing response, or backfill pressure variation.

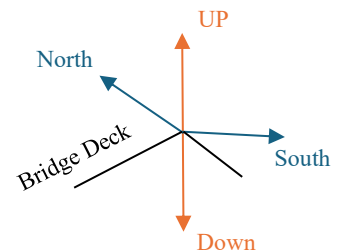
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly non-uniform vertical displacement, indicating significant differential behaviour across the deck. The maximum uplift is observed at the upper LHS region, reaching approximately +20.4 mm, representing the most pronounced vertical response in this configuration.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability and asymmetry, reflecting torsional effects. The central RHS region experiences significant lateral displacement of approximately -11.2 mm, representing the dominant horizontal movement direction

CONCLUSION:

- The bridge deck slab demonstrates a strongly non-symmetric deformation pattern, characterized by large vertical uplift on the LHS and pronounced horizontal displacement toward the RHS. The coexistence of uplift on one side and settlement on the opposite side indicates torsional rotation of the deck slab, rather than uniform vertical translation.